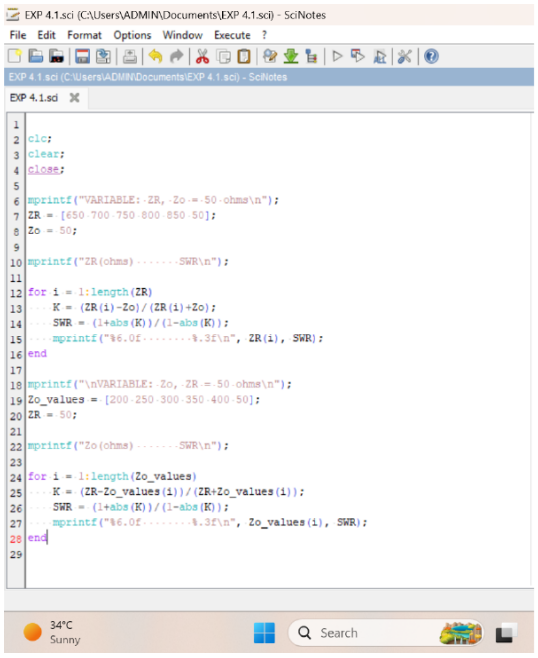
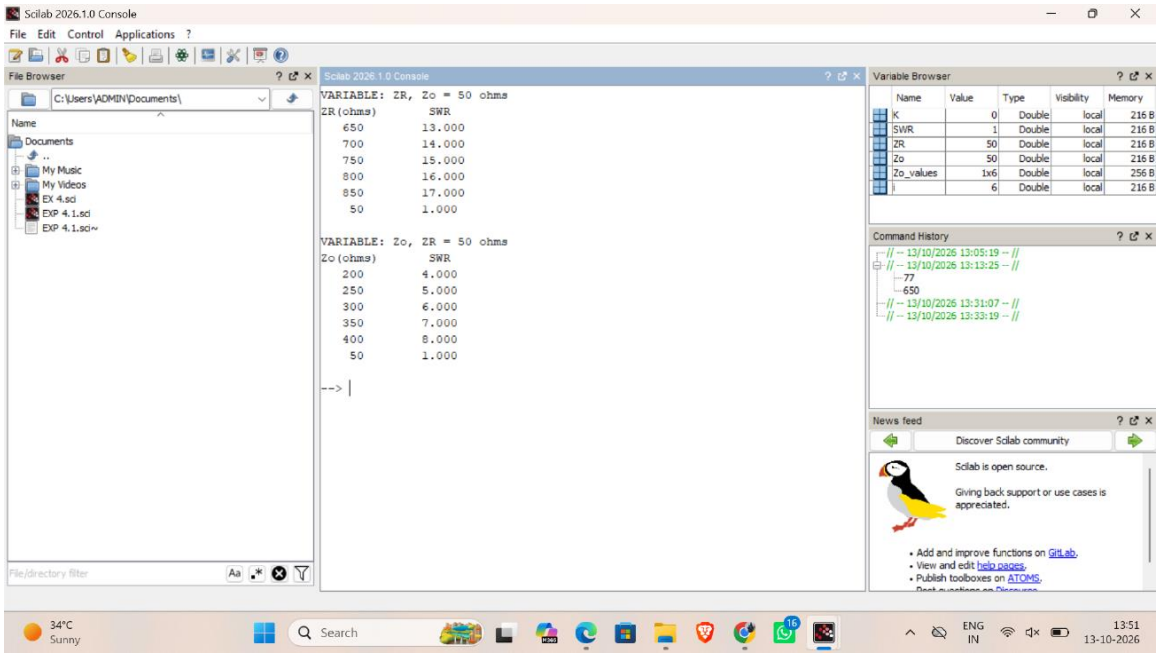
EXNO: 4

CODE:

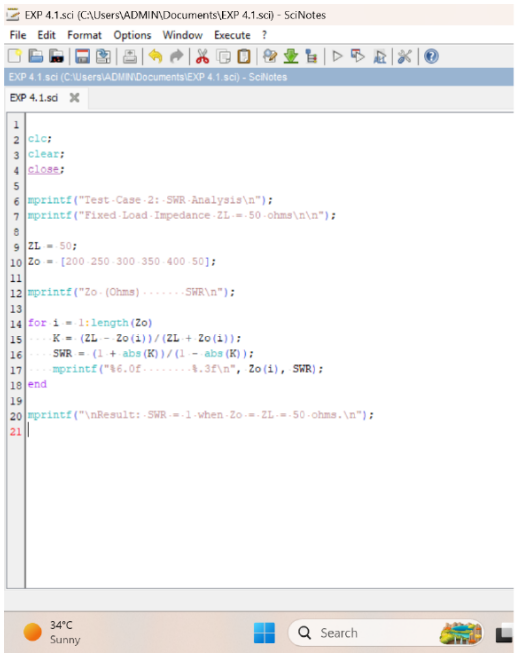


OUTPUT:

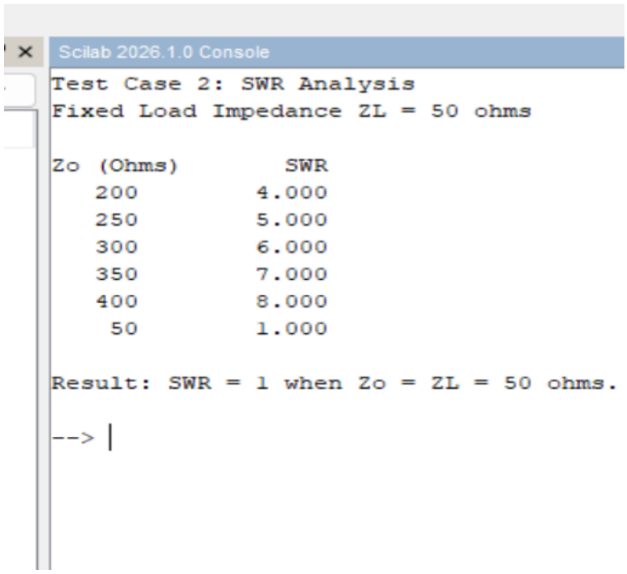


Test Case 2:

CODE:

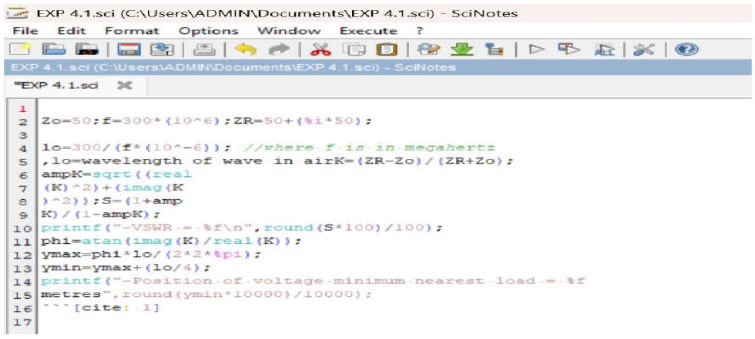


OUTPUT:



EXNO: 5

CODE:



```
Zo=50;f=300*(10^6);ZR=50+(%i*50);

lo=300/(f*(10^-6)); //where f is in megahertz
,lo=wavelength of wave in airK=(ZR-Zo)/(ZR+Zo);
ampK=sqrt((real
(K)^2)+(imag(K
)^2));S=(1+amp
K)/(1-ampK);
printf("-VSWR = %f\n",round(S*100)/100);
phi=atan(imag(K)/real(K));
ymax=phi*lo/(2*2*%pi);
ymin=ymax+(lo/4);
printf("-Position of voltage minimum nearest load = %f
metres",round(ymin*10000)/10000);
```

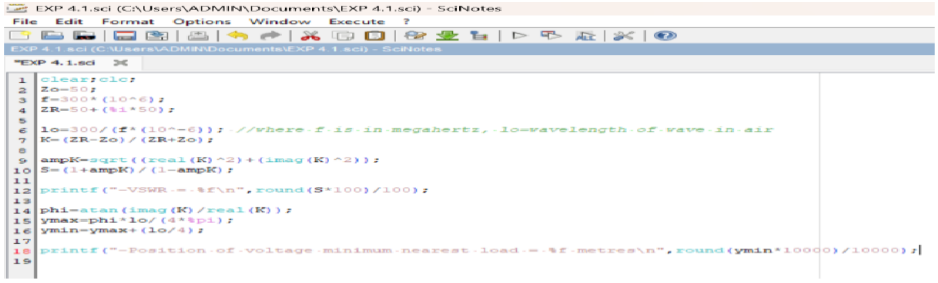
OUTPUT:

```
--> exec('C:\Users\ADMIN\Documents\EXP 4.1.sci', -1)
Variable: ZR Real Part
ZROhm    SWR      Position of Vmin (Simulation & Theoretical)
50       2.62     0.3381
60       2.46     0.3253
70       2.41     0.3133
80       2.44     0.3028

Variable: Zo
ZoOhm    SWR      Position of V min (Simulation & Theoretical)
50       2.62     0.3381
100      2.62     0.1619
150      3.37     0.1936
200      4.27     0.2087
```

Test Case 2:

CODE:



```
clear;clc;
Zo=50;
f=300*(10^6);
ZR=50+(%i*50);

lo=300/(f*(10^-6)); //where f is in megahertz, lo=wavelength of wave in air
K=(ZR-Zo)/(ZR+Zo);

ampK=sqrt((real(K)^2)+(imag(K)^2));
S=(1+ampK)/(1-ampK);

printf("-VSWR = %f\n",round(S*100)/100);

phi=atan(imag(K)/real(K));
ymax=phi*lo/(4*%pi);
ymin=ymax+(lo/4);

printf("-Position of voltage minimum nearest load = %f metres\n",round(ymin*10000)/10000);
```

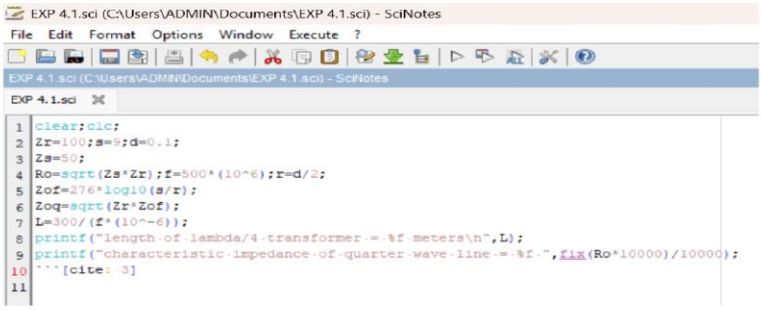
OUTPUT:

```
--> exec('C:\Users\ADMIN\Documents\EXP 4.1.sci', -1)
Test Case 2: Varying Load Impedance
-----------------------------------------------------------
ZL(Ohm)  |K|      Phase(rad)        Position of Vmin (m)
-----------------------------------------------------------
25       0.33     3.14              0.5000
50       0.00     0.00              0.2500
100      0.33     0.00              0.2500
150      0.50     0.00              0.2500
-----------------------------------------------------------

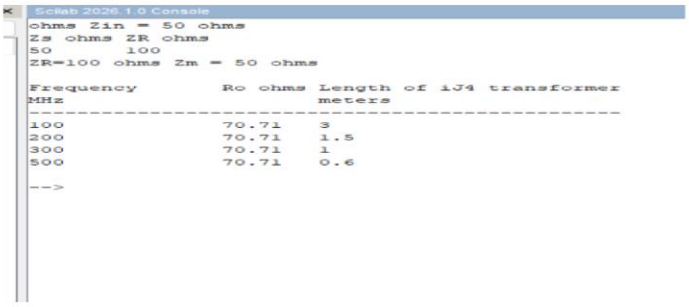
-->
```

EXNO: 6

CODE:

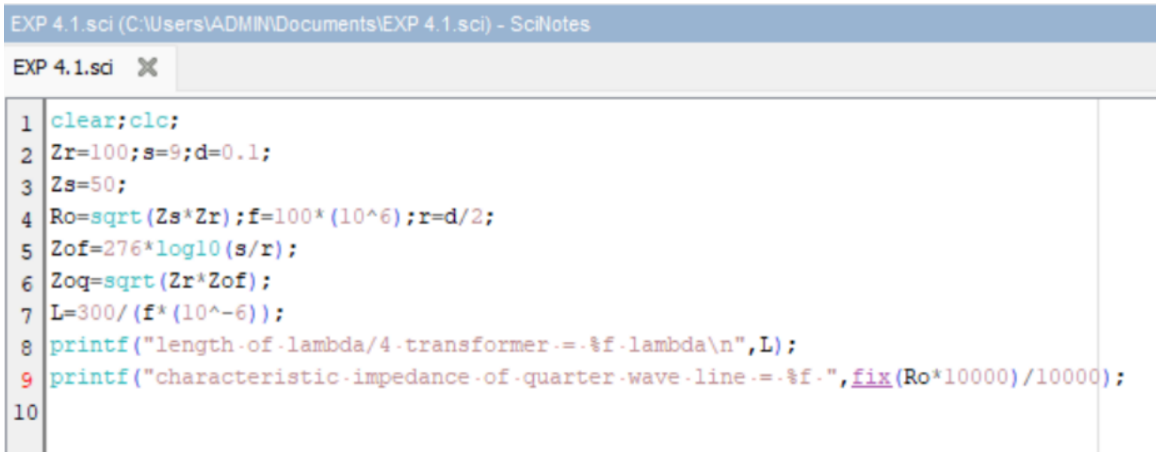


OUTPUT:

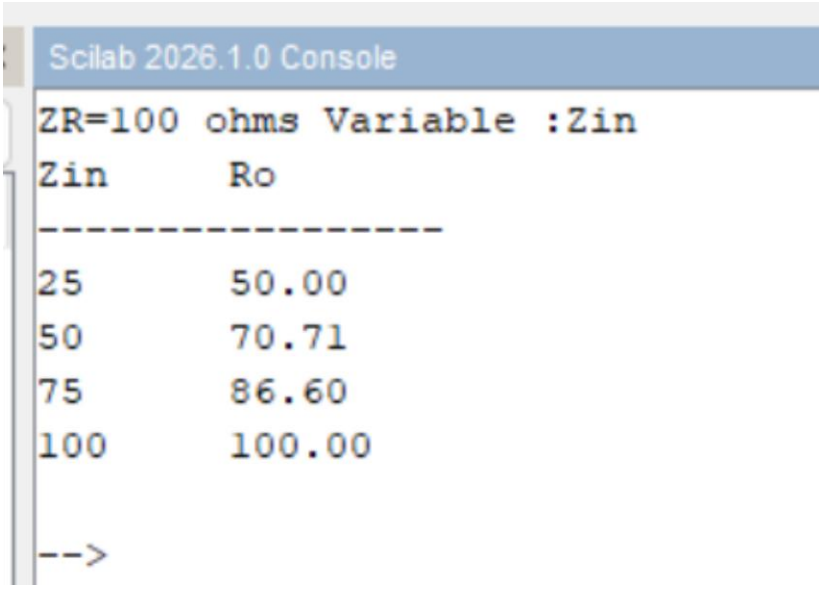


Test Case 2:

CODE:



OUTPUT:


Scilab 2026.1.0 Console
ZR=100 ohms Variable :Zin
Zin Ro
25 50.00
50 70.71
75 86.60
100 100.00
-->